STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

PREPARATORY EXAM – 1 (2026-2027)

Class	**: XII**	**Duration**	**: 2 Hours**
Subject	**: Computer Science (083)**	**Max. Marks**	**: 50**

Section A (14 x 1 = 14 Marks)

1. Which of the following statements about Python exception handling is correct?

a) The except block is executed only when no exception occurs.

b) The finally block can be used to execute statements whether an exception occurs or not.

c) The try block must always contain an else block.

d) A program can have only one except block.

2. What will be the output of the following Python code?

```
try:
    n = int("25")
    print(n + 5)
except ValueError:
    print("Invalid value")
finally:
    print("Completed")
```

a) 25
Completed

b) 30
Completed

c) Invalid value
Completed

d) 30
Completed

3. Which of the following refers to the set of possible values that an attribute can take?

a) Tuple b) Domain c) Cardinality d) Degree

4. Which SQL command is used to modify the structure of an existing table?

a) UPDATE b) ALTER c) MODIFY d) CHANGE

5. Which SQL keyword is used to assign a temporary name to a column in the output of a query?

a) DISTINCT b) AS c) LIKE d) IN

6. Assume that the table faculty contains 8 records. What will be displayed by the following Python statement?

```
cur.execute("SELECT COUNT(*) FROM faculty")
print(cur.fetchone())
```

a) 8 b) (8,) c) [8] d) COUNT(*)

7. Which of the following is the correct way to test whether the value of column phone is not NULL?

a) phone <> NULL

b) phone != NULL

c) phone IS NOT NULL

d) phone NOT NULL

8. Which of the following represents the number of tuples in a relation?

a) Degree b) Attribute c) Cardinality d) Domain

9. Which of the following is a valid MySQL data type for storing a date?

a) DATE b) DATETIMEONLY c) CALENDAR d) DAY

10. What will be the output of the following Python code?

```
try:
    a = [10, 20, 30]
    print(a[1])
except IndexError:
    print("Invalid index")
except Exception:
    print("Something went wrong")
```

a) Something went wrong b) 20 c) 30 d) Invalid index

11. Which Python method is used to retrieve all remaining rows from a MySQL result set?

a) fetchone() b) fetchall() c) fetchrows() d) getall()

12. Which SQL command is used to display the structure of a table?

a) SHOW STRUCTURE student;

b) DISPLAY student;

c) DESCRIBE student;

d) SHOW COLUMNS student;

Q13 and Q14 are Assertion (A) and Reason (R) based questions. Choose the correct option:

a) Both A and R are True and R is the correct explanation for A.

b) Both A and R are True but R is not the correct explanation for A.

c) A is True but R is False.

d) A is False but R is True.

13. Assertion (A): The commit() method is required after executing an INSERT, UPDATE or DELETE operation when changes need to be permanently saved in the database.

Reason (R): commit() permanently saves the changes made during the current database transaction.

14. Assertion (A): An alternate key is a candidate key that has not been selected as the primary key.
Reason (R): A relation can have more than one candidate key, but only one of them is normally selected as the primary key.

Section B (6 x 2 = 12 Marks)

15. A. What is a candidate key? How is an alternate key related to a candidate key?

OR

B. Differentiate between degree and cardinality of a relation with a suitable example.

16. Write a Python program to connect to a MySQL database named schooldb using the following details: Host – localhost, Username – root, Password – school123. After establishing the connection, display "Database Connected Successfully".

17. Predict the output of the following Python code:

```
try:
    x = 10
    y = 0
    print(x / y)
except ZeroDivisionError:
    print("Cannot divide by zero")
except Exception:
    print("Something went wrong")
finally:
    print("End of calculation")
```

18. A. Write suitable MySQL commands to:

I. Display all databases available in the MySQL server.

II. Create a database named STANES.

OR

B. I. Use the database STANES.

II. Display all tables present in the currently selected database.

19. Differentiate between the DELETE and DROP commands in MySQL. Give one suitable example of each.

20. Write a Python function named insert_faculty() to insert a new record into the MySQL table faculty having the columns facty_id, name and salary.

Section C (6 x 4 = 24 Marks)

21. Consider the following table **EMPLOYEE**:

Emp_ID	Emp_Name	Department	Salary
101	Arjun	IT	65000
102	Meena	HR	48000
103	Kavin	IT	72000
104	Riya	Finance	58000
105	Sanjay	HR	52000
106	Divya	IT	61000

Write SQL queries for the following:

I. Display the names and salaries of employees whose salary is greater than 60000.

II. Display the total salary paid by each department.

III. Display the details of employees whose names end with 'a'.

IV. Display the employees in ascending order of their salary.

22. Consider the following table **RESULT**:

Student_ID	Student_Name	Subject	Marks
201	Ananya	Computer Science	92
202	Rahul	Computer Science	78
203	Diya	Computer Science	88
204	Karan	Computer Science	65

Write a Python-MySQL program to update the marks to 72 for the student whose Student_ID is 204.

Use the following connection details:

Database – schooldb, Username – admin, Password – admin@123, Host – localhost. The program should use execute() and commit().

OR

Write a Python-MySQL program to delete the record whose Student_ID is 204 from the RESULT table. After deletion, display the number of rows affected using rowcount.

23. Consider the following table **SPORTS_REG**:

Reg_ID	Player_Name	Sport_Name	Grade	Reg_Fee
501	Aarav	Cricket	11	300
502	Riya	Football	10	200
503	Krish	Cricket	12	300
504	Navya	Badminton	11	250
505	Sameer	Football	9	200
506	Rohit	Athletics	10	150

Write SQL queries for the following:

I. Display the number of players registered for each sport.

II. Display the names of players whose names contain the letters 'ri'.

III. Display the highest registration fee.

IV. Display the sport name and the average registration fee for each sport.

24. Predict the output of the following SQL queries of the same table SPORTS_REG:

```
I. SELECT Player_Name, Sport_Name FROM SPORTS_REG
   WHERE Grade IN (10, 11) ORDER BY Reg_ID;

II. SELECT Sport_Name, COUNT(*) FROM SPORTS_REG
    GROUP BY Sport_Name ORDER BY Sport_Name;

III. SELECT Player_Name FROM SPORTS_REG
     WHERE Player_Name LIKE 'R%' ORDER BY Player_Name;

IV. SELECT SUM(Reg_Fee) FROM SPORTS_REG
    WHERE Grade BETWEEN 10 AND 11;
```

25. Consider the following two tables:

Table: EVENT

Event_ID	Event_Name	Event_Date	Venue
101	Science Expo	2026-01-15	Main Hall
102	Coding Contest	2026-02-10	Lab Block
103	Robotics Meet	2026-03-05	Main Hall
104	Mathematics Fest	2026-04-12	Activity Hall

Table: REGISTRATION

Reg_ID	Event_ID	Student_Name
1	101	Ananya
2	102	Rahul
3	101	Kavin
4	103	Diya
5	104	Arjun

Write SQL queries for the following:

I. Update the venue of "Coding Contest" to "Computer Lab".

II. Display the details of events scheduled after 2026-02-01.

III. Display the names of students who have registered for events conducted in "Main Hall" using an appropriate join.

IV. Display the Cartesian product of the EVENT and REGISTRATION tables.

26. Write a Python-MySQL program to connect to a database named College, retrieve all records from the table Faculty using fetchall(), and display each record. The program must handle any exception that may occur during the process.

Use the following credentials:

Host – localhost

Username – admin

Password – admin@123.